



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

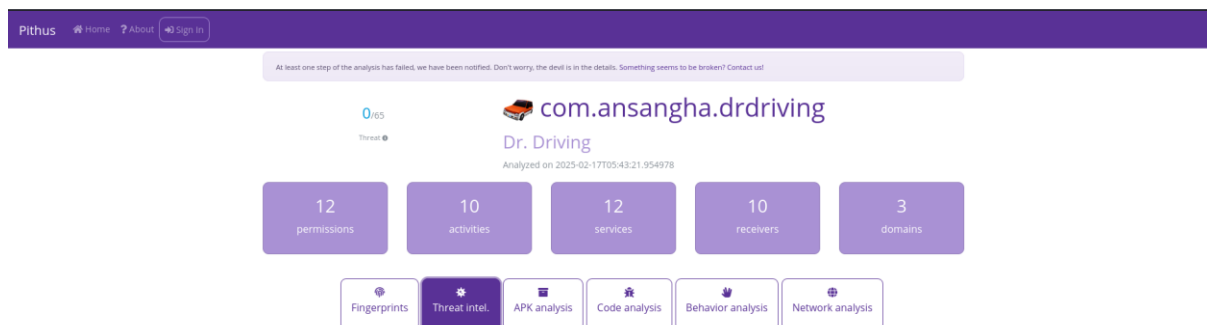
EXPERIMENT 3

Aim: Android Malware analysis using Pithus.

Theory:

Pithus is an open-source, web-based Android malware analysis platform that provides automatic static analysis of APK files to detect potential threats, vulnerabilities, and malicious behavior. It scans APKs for dangerous permissions, suspicious API calls, obfuscation techniques, and embedded payloads, helping security researchers and penetration testers identify malware efficiently. Unlike traditional reverse engineering tools, Pithus requires no installation, making it a fast and accessible solution for Android security analysis. Users can simply upload an APK on [Pithus.org](https://pithus.org) to generate a detailed report, making it a valuable tool for malware detection and forensic investigations.

Screenshots:

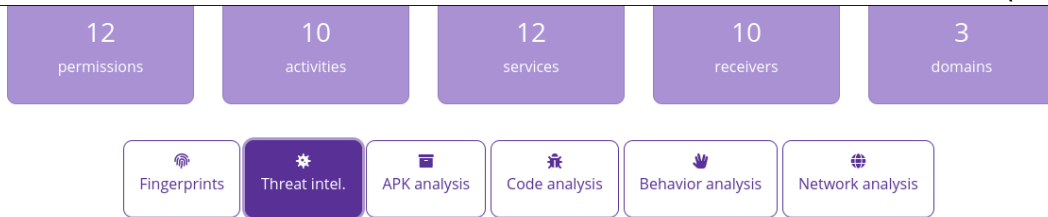


Threats:



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Sample timeline

Certificate valid not before	Jan. 24, 2017, 1:11 p.m.
Oldest file found in APK	Sept. 2, 2024, 3:52 p.m.
Latest file found in APK	Sept. 2, 2024, 3:57 p.m.
First submission on VT	Sept. 2, 2024, 4:19 p.m.
Last submission on VT	Feb. 17, 2025, 12:55 a.m.
Upload on Pithus	Feb. 17, 2025, 5:43 a.m.
Certificate valid not after	June 10, 2044, 1:11 p.m.

VirusTotal

Samples are analysed on upload and are not analysed afterwards again. Make sure to review the status of the analysis on VirusTotal to be sure to have the most up-to-date conclusion.

Score

APK Details:

APK details

Information computed with AndroGuard and Pithus.

Package	com.ansangha.drdriving	
App name	Dr. Driving	
Version name	1.72	
Version code	72	
SDK	21 - 34	
UAID	dfd3e8838174d7c3dce81a65af46d9c3d7d1e65d	
Signature	Signature V1 Signature V2 Signature V3	
Frosting	Not frosted Blocks found within V2 signature: 0x7109871a: Unknown 0xf05368c0: Unknown 0x42726577: Verity padding	

Certificate details

Information computed with AndroGuard.

MD5	3ada5333c68fba2dc8c7816ead644a47	
SHA1	fd7a95648ff46fb1acab854acfe1bf97ecf92346	



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Certificate details

Information computed with [AndroGuard](#).

MD5	3ada5333c68fba2dcdc7816ead644a47	
SHA1	fd7a95648ff46fb1acab854acfe1bf97ecf92346	
SHA256	1cbd2d0548bb11026d9ab4dbad4a2c9405ce796235d4003bcf0a0aed8bfe0c98	
Issuer	Email Address: a1@tr4oo3r34r34sdfd.com, Common Name: A1, Organizational Unit: A1, Organization: A1, Locality: Lazyland, State/Province: Lazyland, Country: RU	
Not before	2017-01-24T13:11:10+00:00	
Not after	2044-06-10T13:11:10+00:00	

File Analysis

Information computed with [MobSF](#).

Findings	Files
Certificate/Key files hardcoded inside the app.	META-INF/services/io.grpc.ManagedChannelProvider

Manifest analysis

Information computed with [MobSF](#).

Medium	Application Data can be Backed up [android:allowBackup=true] This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.
High	Service (com.google.android.gms.auth.api.signin.RevocationBoundService) is Protected by a permission, but the protection level of the permission should be checked. Permission: com.google.android.gms.auth.api.signin.permission.REVOCATION_NOTIFICATION [android:exported=true] A Service is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.
High	Activity (com.google.android.gms.games.internal.v2.appshortcuts.PlayGamesAppShortcutsActivity) Is not Protected. [android:exported=true] An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.
High	Service (androidx.work.impl.background.systemjob.SystemJobService) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.BIND_JOB_SERVICE [android:exported=true] A Service is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.
High	Broadcast Receiver (androidx.work.impl.diagnostics.DiagnosticsReceiver) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.DUMP [android:exported=true] A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.



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Main Activity

Information computed with AndroGuard.

com.ansangha.drdriving.DrDrivingActivity



Activities

Information computed with AndroGuard.

com.ansangha.drdriving.DrDrivingActivity
com.google.android.gms.ads.AdActivity
com.android.billingclient.api.ProxyBillingActivity
com.android.billingclient.api.ProxyBillingActivityV2
com.google.android.gms.auth.api.signin.internal.SignInHubActivity
com.google.android.gms.games.internal.v2.resolution.GamesResolutionActivity
com.google.android.gms.games.internal.v2.appshortcuts.PlayGamesAppShortcutsActivity
com.google.android.gms.common.api.GoogleApiActivity
com.google.android.gms.ads.OutOfContextTestingActivity
com.google.android.gms.ads.NotificationHandlerActivity



Receivers

Information computed with AndroGuard.

com.google.android.datatransport.runtime.scheduling.jobscheduling.AlarmManagerSchedulerBroadcastReceiver
com.google.android.gms.measurement.AppMeasurementReceiver
androidx.work.impl.utils.ForceStopRunnable\$BroadcastReceiver
androidx.work.impl.background.systemalarm.ConstraintProxy\$BatteryChargingProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$BatteryNotLowProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$StorageNotLowProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$NetworkStateProxy
androidx.work.impl.background.systemalarm.RescheduleReceiver



Receivers

Information computed with AndroGuard.

com.google.android.datatransport.runtime.scheduling.jobscheduling.AlarmManagerSchedulerBroadcastReceiver
com.google.android.gms.measurement.AppMeasurementReceiver
androidx.work.impl.utils.ForceStopRunnable\$BroadcastReceiver
androidx.work.impl.background.systemalarm.ConstraintProxy\$BatteryChargingProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$BatteryNotLowProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$StorageNotLowProxy
androidx.work.impl.background.systemalarm.ConstraintProxy\$NetworkStateProxy
androidx.work.impl.background.systemalarm.RescheduleReceiver
androidx.work.impl.background.systemalarm.ConstraintProxyUpdateReceiver
androidx.work.impl.diagnostics.DiagnosticsReceiver

Services

Information computed with AndroGuard.

com.google.android.gms.auth.api.signin.RevocationBoundService
com.google.firebase.components.ComponentDiscoveryService
com.google.firebase.sessions.SessionLifecycleService
com.google.android.datatransport.runtime.backends.TransportBackendDiscovery
com.google.android.datatransport.runtime.scheduling.jobscheduling.JobInfoSchedulerService
com.google.android.gms.measurement.AppMeasurementService
com.google.android.gms.measurement.AppMeasurementJobService
com.google.android.gms.ads.AdService
androidx.work.impl.background.systemalarm.SystemAlarmService
androidx.work.impl.background.systemjob.SystemJobService
androidx.work.impl.foreground.SystemForegroundService
androidx.room.MultiInstanceInvalidationService



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Code Analysis:

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NIAP analysis

Information computed with MobSF.

FCS_RBG_EXT.1.1	The application invoke platform-provided DRBG functionality for its cryptographic operations. Random Bit Generation Services
FCS_STO_EXT.1.1	The application does not store any credentials to non-volatile memory. Storage of Credentials
FCS_CKM_EXT.1.1	The application implement asymmetric key generation. Cryptographic Key Generation Services
FDP_DEC_EXT.1.1	The application has access to [network connectivity]. Access to Platform Resources
FDP_DEC_EXT.1.2	The application has access to no sensitive information repositories. Access to Platform Resources
FDP_NET_EXT.1.1	The application has user/application initiated network communications. Network Communications
FDP_DAR_EXT.1.1	The application implement functionality to encrypt sensitive data in non-volatile memory. Encryption Of Sensitive Application Data
FMT_MEC_EXT.1.1	The application invoke the mechanisms recommended by the platform vendor for storing and setting configuration options. Supported Configuration Mechanism
FTP_DIT_EXT.1.1	The application does encrypt some transmitted data with HTTPS/TLS/SSH between itself and another trusted IT product. Protection of Data in Transit
FCS_RBG_EXT.2.1 FCS_RBG_EXT.2.2	The application perform all deterministic random bit generation (DRBG) services in accordance with NIST Special Publication 800-90A using Hash_DRBG. The deterministic RBG is seeded by an entropy source that accumulates entropy from a platform-based DRBG and a software-based noise source, with a minimum of 256 bits of entropy at least equal to the greatest security strength (according to NIST SP 800-57) of the keys and hashes that it will generate. Random Bit Generation from Application
FCS_COP.1.1(1)	The application perform encryption/decryption in accordance with a specified cryptographic algorithm AES-CBC (as defined in NIST SP 800-38A) mode or AES-GCM (as defined in NIST SP 800-38D) and cryptographic key sizes 256-bit/128-bit.

Code analysis

Information computed with MobSF.

Medium CVSS:7.5 	🔗 The App uses an Insecure Random Number Generator. MASVS: MSTG-CRYPTO-6 CWE-330 Use of Insufficiently Random Values M5: Insufficient Cryptography
Medium CVSS:7.4 	🔗 Files may contain hardcoded sensitive information like usernames, passwords, keys etc. MASVS: MSTG-STORAGE-14 CWE-312 Cleartext Storage of Sensitive Information M9: Reverse Engineering
Low CVSS:7.5 	🔗 The App logs information. Sensitive information should never be logged. MASVS: MSTG-STORAGE-3 CWE-532 Insertion of Sensitive Information into Log File
Medium CVSS:5.9 	🔗 App uses SQLite Database and execute raw SQL query. Untrusted user input in raw SQL queries can cause SQL Injection. Also sensitive information should be encrypted and written to the database. CWE-89 Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection') M7: Client Code Quality
Medium CVSS:5.9 	🔗 SHA-1 is a weak hash known to have hash collisions. MASVS: MSTG-CRYPTO-4 CWE-327 Use of a Broken or Risky Cryptographic Algorithm M5: Insufficient Cryptography
High CVSS:5.5 	🔗 App can read/write to External Storage. Any App can read data written to External Storage. MASVS: MSTG-STORAGE-2 CWE-276 Incorrect Default Permissions M2: Insecure Data Storage

Behaviour Analysis:



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Permissions analysis

Information computed with MobSF.

Low	android.permission.INTERNET	full Internet access Allows an application to create network sockets.	
Low	android.permission.ACCESS_NETWORK_STATE	view network status Allows an application to view the status of all networks.	
Low	android.permission.WAKE_LOCK	prevent phone from sleeping Allows an application to prevent the phone from going to sleep.	
Low	android.permission.FOREGROUND_SERVICE	Allows a regular application to use Service.startForeground.	
	com.android.vending.BILLING	Unknown permission Unknown permission from android reference	
	com.google.android.providers.gsf.permission.READ_GSERVICES	Unknown permission Unknown permission from android reference	
	com.google.android.gms.permission.AD_ID	Unknown permission Unknown permission from android reference	
	android.permission.ACCESS_AD SERVICES_ATTRIBUTION	Unknown permission Unknown permission from android reference	
	android.permission.ACCESS_AD SERVICES_AD_ID	Unknown permission Unknown permission from android reference	
	com.google.android.finsky.permission.BIND_GET_INSTALL_REFERRER_SERVICE	Unknown permission Unknown permission from android reference	
	android.permission.ACCESS_AD SERVICES_TOPICS	Unknown permission Unknown permission from android reference	
	com.ansangha.drdriving.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION	Unknown permission Unknown permission from android reference	



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Tracking analysis

Information computed with Exodus-core.

Google AdMob	https://reports.exodus-privacy.eu.org/fr/trackers/312
Google CrashLytics	https://reports.exodus-privacy.eu.org/fr/trackers/27
Google Firebase Analytics	https://reports.exodus-privacy.eu.org/fr/trackers/49

Threat analysis

Information computed with Quark-Engine.

Confidence: 100%	Check if the network is connected
Confidence: 100%	Load external class
Confidence: 100%	Query the current data network type
Confidence: 100%	Implicit Intent(view a web page, make a phone call, etc.)
Confidence: 100%	Find a method from given class name, usually for reflection
Confidence: 100%	Check the active network type
Confidence: 100%	Read data and put it into a buffer stream
Confidence: 100%	Put the compressed bitmap data into JSON object
Confidence: 100%	Method reflection
Confidence:	Connect to a URL and read data from it



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Behavior analysis

Information computed with MobSF.

Base64 decode
Base64 encode
Certificate handling
Get system service
Http connection
Https connection
Inter process communication
Java reflection
Local file i/o operations
Message digest
Starting activity
Starting service
Tcp socket

Network Analysis:

Domains analysis

Information computed with MobSF.

US	crucial-bonsai-821.firebaseio.com	🔍 📄 G	34.120.206.254	📄 G
US	firebase.google.com	🔍 📄 G	142.250.185.206	📄 G
US	github.com	🔍 📄 G	140.82.121.3	📄 G

URL analysis

Information computed with MobSF.

https://firebase.google.com/docs/database/ios/structure-data#best_practices_for_data_structure https://firebase.google.com/docs/database/android/retrieve-data#filtering_data https://github.com/firebase/firebase-android-sdk Defined in n2/c.java	📄
https://firebase.google.com/docs/database/ios/structure-data#best_practices_for_data_structure https://firebase.google.com/docs/database/android/retrieve-data#filtering_data https://github.com/firebase/firebase-android-sdk Defined in n2/c.java	📄
https://firebase.google.com/docs/database/ios/structure-data#best_practices_for_data_structure https://firebase.google.com/docs/database/android/retrieve-data#filtering_data https://github.com/firebase/firebase-android-sdk Defined in n2/c.java	📄
https://firebase.google.com/docs/crashlytics/get-started?platform=android#add-plugin Defined in x1/q.java	📄
https://crucial-bonsai-821.firebaseio.com Defined in Android String Resource	📄



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Conclusion:

Thus we successfully scanned Dr. Driving game APK for malware using Pithus on web browser in kali. It provided us with a detailed report consisting of fingerprints, threats, domains, urls, analysis, code analysis and many more things.